

BACK OF THE ENVELOPE SYSTEM DESIGN CALCULATIONS

APPROXIMATION TABLE

Storage Unit	Approximate Value
1 KB	$\approx 1,000$
1 MB	$\approx 1,000,000$
1 GB	$\approx 10,000,000$
1 TB	$\approx 10,000,0000$
1 PB	$\approx 1,000,000,00,000$

— THINKING ESTIMATES —

- Latency: ms = microseconds / 1000
- Read / Write Ratio: Eg, 80% Reads, 20% Writes
- Replication Factor: 2x or 3x copies of data
- Cache Hit Ratio: Eg, 90% Hit, 10% Miss

NETWORK SPEEDS —

- 1 Mbps = 0.125 MBps
- 10 Mbps = 1.25 MBps
- 1 Gbps = 125 MBps
- 10 Gbps = 1,250 MBps



KEY FORMULAS

- $\text{QPS} = \text{Queries Per Second}$
- $\text{Data Per Day} = \text{QPS} \times \text{Data Size} \times 86,400$
- $\text{Throughput} = \text{Requests Per Second} \times \text{Data Size}$
- $\text{Bandwidth} = \text{Throughput} \times 8$ (for bits)
- $\text{Storage} = \text{Traffic per Day} \times \text{Retention Days}$
- $\text{Memory} = \text{QPS} \times \text{Session Size} \times \text{Duration}$

TIME CONVERSIONS —

- 1 Second ≈ 1 QPS
- 1 Minute ≈ 60 Seconds
- 1 Hour = 3,600 Seconds
- 1 Day = 86,400 Seconds
- 1 Month = 30 Days ($\sim 2.6\text{M}$ Secs)



— ORDER OF MAGNITUDE —

- 1K $\approx 1,000$ (Thousand)
- 1M $\approx 1,000,000$ (Million)
- 1B $\approx 1,000,000,000$ (Billion)
- 1T $\approx 1,000,000,000,000$ (Trillion)



— CONSIDERATIONS —

- ✓ Traffic Estimates
- ✓ Storage & Database
- ✓ Latency Requirements
- ✓ Consistency Model
- ✓ Scalability & Load Balancing
- ✓ Fault Tolerance
- ✓ CDN & Caching

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- ✓ Cost & Constraints



SYSTEM DESIGN



BACK-OF-THE-ENVELOPE CHEATSHEET

Estimates & Formulas to Crack Any System Design Interview

1. TIME & BASIC MATH

1 sec = 1,000 ms
1 min = 60 sec
1 hour = 3,600 sec
1 day = 86,400 sec
1 year $\approx 3.15 \times 10^7$ sec



Rules of Thumb

- Round up & use powers of 10
- Assume 20–30% overhead
- Prefer Fermi Estimation (break problem into steps)

2. DATA UNITS (Decimal)

Unit	Value	In Bytes
1 KB	$= 10^3$	$= 1,000$ B
1 MB	$= 10^6$	$= 1,000,000$ B
1 GB	$= 10^9$	$= 1,000,000,000$ B
1 TB	$= 10^{12}$	$= 1,000,000,000,000$ B
1 PB	$= 10^{15}$	$= 1,000,000,000,000,000$ B

Binary (Sometimes Used)

1 KiB	$= 2^{10}$	$= 1,024$ B
1 MiB	$= 2^{20}$	$= 1,048,576$ B
1 GiB	$= 2^{30}$	$= 1,073,741,824$ B

Interview Tip: Use Decimal unless asked.

3. NETWORK & LATENCY

Typical Latency

Intra-DC / Same Rack	0.1 – 0.2 ms
Same DC	0.5 – 1 ms
Cross Region	20 – 80 ms
Inter-Continent	100 – 200 ms



Bandwidth (Full Duplex)

- 1 Gbps $= 10^9$ bps ≈ 125 MB/s
- 10 Gbps ≈ 1.25 GB/s
- 100 Gbps ≈ 12.5 GB/s

4. STORAGE ESTIMATION

Total Storage =

Write QPS $\times$ Write Size (B)
 $\times$ Retention (sec) $\times$ Replication Factor



Example:

- QPS = 1,000
- Write Size = 1 KB
- Retention = 30 days
- Replication = 3

Storage $= 1,000 \times 1\text{KB} \times 2.59 \times 10^6 \times 3$
 $= 7.77 \times 10^{12}$ B ≈ 7.8 TB

5. CAPACITY PLANNING



Metric Formula

Requests / Sec	$= \text{DAU} \times \text{Avg Requests per User}$ $= 86,400$
Read QPS	$= \text{Total QPS} \times \text{Read \%}$
Write QPS	$= \text{Total QPS} \times \text{Write \%}$
Data In / Sec	$= \text{Write QPS} \times \text{Write Size}$
Data Out / Sec	$= \text{Read QPS} \times \text{Read Size}$

Rule of Thumb

- Plan for Peak = 2–3 $\times$ Average
- Leave 30–50% Headroom

6. DATABASE ESTIMATION



Key-Value Store

- Item Size = Key + Value + Metadata
- Total Data = Item Size $\times$ Number of Items
- Add 20–30% overhead

Relational DB

- Add Index Size $\approx$ 20–30% of Data
- Add Logs, Backups, Overhead $\approx$ 30%

Connection Pool Size

$\approx (\text{Core Count} \times 2) + \text{Effective Spindle Count}$

7. CACHE ESTIMATION

Hit Ratio Benefit



New Latency = Hit Ratio $\times$ Cache Latency
 $+ (1 - \text{Hit Ratio}) \times \text{DB Latency}$

Example:

- Hit Ratio = 90%
- Cache Latency = 1 ms
- DB Latency = 10 ms

New Latency $= 0.9 \times 1 + 0.1 \times 10 = 1.9$ ms

8. QUEUE & WORKER POOL



Required Workers =

$\frac{\text{Incoming Task Rate} \times \text{Avg Task Time}}{\text{Target CPU Utilization}}$

- Keep CPU Utilization $\leq$ 70%
- Use Queue to absorb spikes

9. CDN & TRAFFIC



CDN Offload

Origin Traffic = Total Traffic $\times (1 - \text{Cache Hit Ratio})$

Example: 1,000 RPS, 80% Hit Ratio

Origin Traffic $= 1,000 \times 0.2 = 200$ RPS

10. COMMON ASSUMPTIONS

Area	Assumption
Users	10^6 MAU = Large Consumer App
Traffic	10–20% of MAU active daily
Requests/User/Day	10–100 (varies by product)
Object Size	1–100 KB (default 10 KB)
Read:Write Ratio	80:20 (typical)
Data Growth	1–2 $\times$ per year
Replication Factor	3 (common)



11. FERMI ESTIMATION STEPS

1. Define the Goal
2. Make an Assumption
3. Break Down the Problem
4. Calculate & Adjust
5. Sanity Check



12. QUICK REFERENCE FORMULAS

- Throughput (RPS) = $\text{Concurrency} / \text{Latency (sec)}$
- Concurrency = $\text{RPS} \times \text{Latency}$
- Data Transfer Time = $\text{Data Size} / \text{Bandwidth}$
- CPU Cores Needed = $\text{Throughput} \times \text{CPU Time} / \text{Core}$
- Storage with Replication = $\text{Raw Data} \times \text{Replication Factor}$
- Monthly Active Storage Cost = $\text{Storage} \times \text{Cost per GB}$

Remember: Communicate your assumptions, think out loud, and iterate with the interviewer.